

```
1 import java.util.*;
2 class A1Z26
3 {
4     String s;
5     A1Z26()
6     {
7         s="";
8     }
9
10    void choice()
11    {
12        Scanner sc=new Scanner(System.in);
13        int ch;
14        System.out.println("1.Encode");
15        System.out.println("2.Decode");
16        System.out.print("Enter your choice:");
17        ch=sc.nextInt();//accepting choice
18        switch(ch)
19        {
20            case 1:
21                System.out.println("Enter the sentence to be encoded:");
22                s=sc.nextLine();//will not be excuted by Scanner Class
23                s=sc.nextLine();//taking input
24                encode(s.toLowerCase().trim());//calling function
25                break;
26            case 2:
27                System.out.println("Enter the sentence to be decoded:");
28                s=sc.nextLine();//will not be excuted by Scanner Class
29                s=sc.nextLine();//taking input
30                decode(s.trim());//calling function
31                break;
32            default:
33                System.out.println("Invalid Input");
34        }
35    }
36
37    void encode(String s)
38    {
39        System.out.println("The Encoded Message:");
40        for(int i=0;i<s.length();i++)
41        {
42            if(Character.isLetter(s.charAt(i)))
43            {
44                if(i==s.length()-1)
45                    System.out.print(s.charAt(i)-96);
46                else
47                    if ( s.charAt(i+1) !=' ')
48                        System.out.print(s.charAt(i)-96+"-");
49                else
50                    System.out.print(s.charAt(i)-96);
51            }
52        }
53    }
54 }
```

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51         else
52         if ( s.charAt(i) == ' ')
53             System.out.print(s.charAt(i));
54         else
55             System.out.print(s.charAt(i)+"-");
56     }
57 }
58
59 void decode(String s)
60 {
61     System.out.println("The Decoded Message:");
62     StringTokenizer ob=new StringTokenizer(s);
63     StringTokenizer ob1;
64     String ch,ch1;
65     while(ob.hasMoreTokens())
66     {
67         ch=ob.nextToken();
68         ob1=new StringTokenizer(ch,"-");
69         while(ob1.hasMoreTokens())
70         {
71             ch1=ob1.nextToken();
72             try
73             {
74                 System.out.print((char)(Integer.parseInt(ch1)+96)); //decoding the
message
75             }
76             catch(NumberFormatException e)
77             {
78                 System.out.print(ch1);
79             }
80         }
81         System.out.print(" ");
82     }
83 }
84
85 public static void main()
86 {
87     A1Z26 obj=new A1Z26();
88     obj.choice();//calling function
89 }
90 }
91
```